```
> alpha := proc(m, n)
    if n = 1 then
      (-1)^m;
    elif n = 2 then
      (-1)^(m + 1) * m^2;
    else
      (-1)^(m+n-1)·2^(n-2)·m·2^(3-2n)·(m+n-2)! / (pochhammer(1/2, n-1)·(m-n+1)!)
    end if;
  end proc:
```

$$\frac{(-1)^{m+n-1}\cdot 2^{n-2}\cdot m\cdot 2^{3-2n}\cdot (m+n-2)!}{\mathrm{pochhammer}\left(\frac{1}{2}, n-1\right)\cdot (m-n+1)!}$$

```
> V := unapply(sum(alpha(m, n) * (exp(I*y) / (I*y)^n + (-1)^(m + n) * exp(-I*y) / (I*y)^n), n = 1 .. m + 1), m, y);
```

$$V := (m, y) \mapsto -\frac{\mathrm{I}\cdot e^{\mathrm{I}\cdot y}\cdot (-1)^m\cdot \mathrm{hypergeom}\left([1, m, -m], \left[\frac{1}{2}\right], \frac{-\frac{\mathrm{I}}{2}}{y}\right)}{y} - \frac{\mathrm{I}\cdot e^{-\mathrm{I}\cdot y}\cdot (-1)^m\cdot (-1)^{m+1}\cdot \mathrm{hypergeom}\left([1, m, -m], \left[\frac{1}{2}\right], \frac{\frac{\mathrm{I}}{2}}{y}\right)}{y} \quad (1)$$

```
> tm := proc(m, y)
    piecewise(m = 0, 2 sin(y) / y, m = 1, -V(m, y), V(m, y));
  end proc:
```

```
>
```

```
> X := (n, y)→simplify(int(ChebyshevT(n, x)·exp(-I·x·y), x = -1 ..1)) :
```

```
>
```

```
> Y := unapply(simplify(tm(m, y) / sqrt(4 (-1)^m Pi (2 m^2 - 1) / (4 m^2 - 1))), m, y) :
    # Normalized Fourier transformed Type-I Chebyshev Polynomials
```

```
> Matrix(map(simplify, [seq([(X(m, y)), (tm(m, y))], m = 0 ..4)]))
```

$$\left[\left[\frac{2\sin(y)}{y}, \frac{2\sin(y)}{y}\right], \left[\frac{-2\,\mathrm{I}\,(-y\cos(y)+\sin(y))}{y^2}, \frac{-2\,\mathrm{I}\,(y\cos(y)-\sin(y))}{y^2}\right], \left[\frac{2\sin(y)\,y^2+8\,y\cos(y)-8\sin(y)}{y^3}, \frac{2\sin(y)\,y^2+8\,y\cos(y)-8\sin(y)}{y^3}\right], \quad (2)$$

$$\left[\frac{2\,\mathrm{I}\left(\cos(y)\,y^3 - 9\sin(y)\,y^2 - 24\,y\cos(y) + 24\sin(y)\right)}{y^4},\ \frac{2\,\mathrm{I}\left(\cos(y)\,y^3 - 9\sin(y)\,y^2 - 24\,y\cos(y) + 24\sin(y)\right)}{y^4}\right],$$

$$\left[\frac{\left(2\,y^4 - 160\,y^2 + 384\right)\sin(y) + 32\,y\left(y^2 - 12\right)\cos(y)}{y^5},\ \frac{\left(2\,y^4 - 160\,y^2 + 384\right)\sin(y) + 32\,y\left(y^2 - 12\right)\cos(y)}{y^5}\right]\Bigg]$$

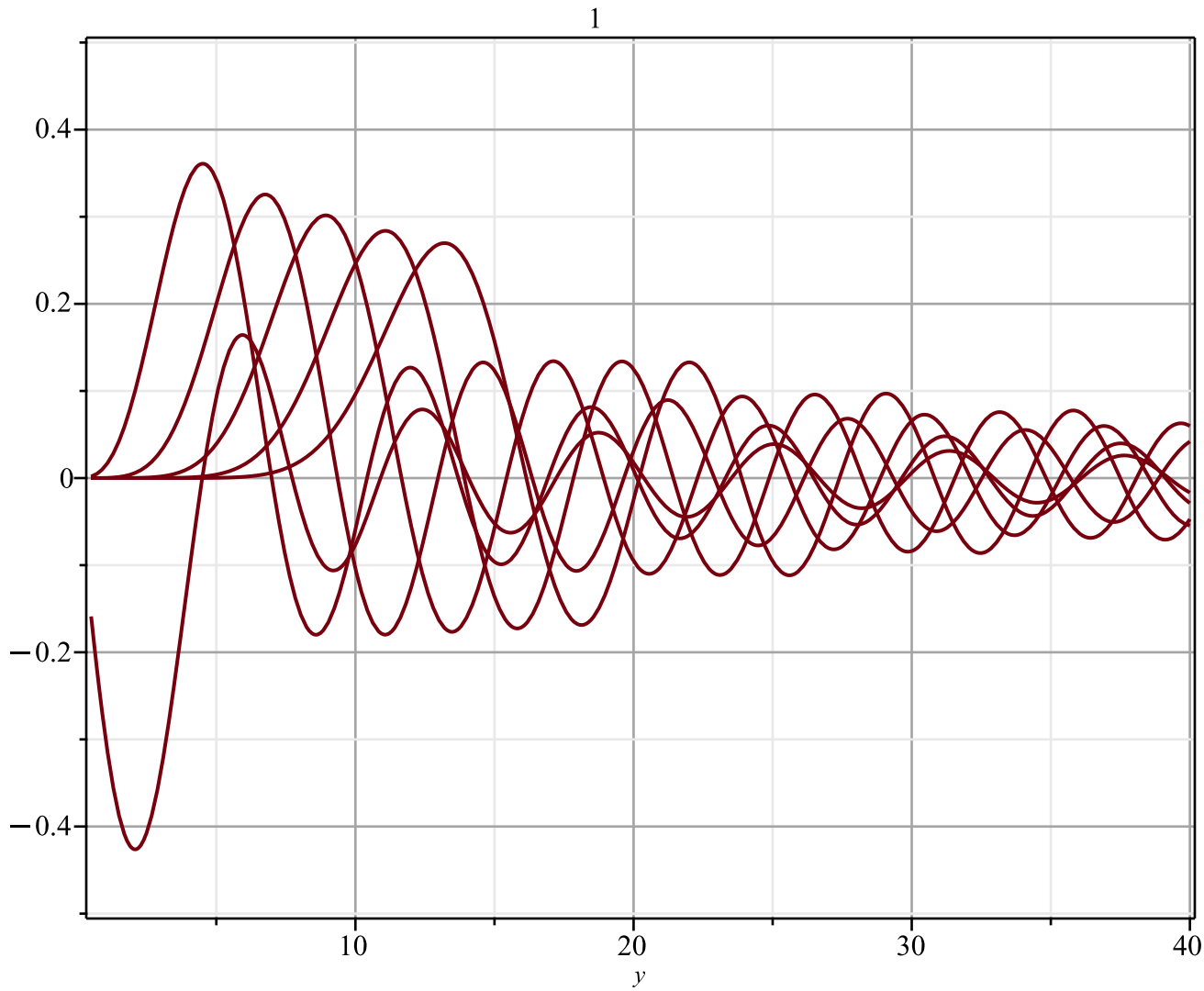
```
>
>
> f := unapply(int(BesselJ(0, y)·exp(−I·x·y), y = 0 ..infinity) , x) assuming x ≠ 0
```

$$f := x \mapsto \frac{1}{\sqrt{-x^2 + 1}} \tag{3}$$

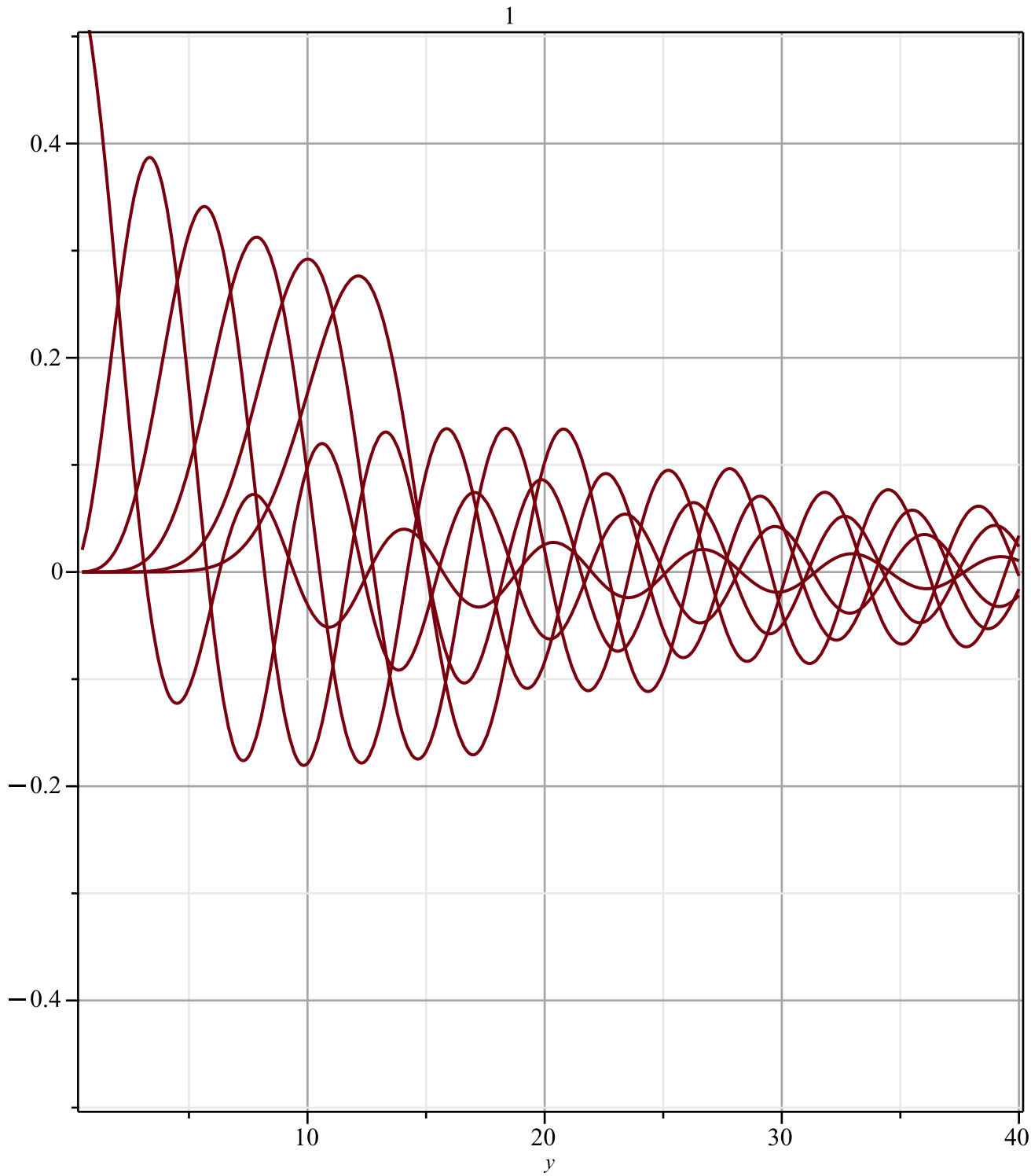
```
> # Gram-Schmidt process for the Fourier transforms of Chebyshev polynomials
  GramSchmidt := proc(vecList)
  local n, m, inner, projection, u, v, orthoList;
  orthoList := [ ];
  for n from 1 to numelems(vecList) do
     v := vecList[n];
     projection := 0;
     for m from 1 to n−1 do
        u := orthoList[m];
        inner := int(u * v, y = −infinity .. infinity);
        projection := projection + inner * u;
     end do;
     u := simplify(v − projection);
     u := simplify(u / sqrt(int(u * conjugate(u), y = −infinity .. infinity)));
     orthoList := [op(orthoList), u];
  end do;
  return orthoList;
   end proc:
> # Generate the functions
  funcList := map(simplify, [seq(Y(n, y), n = 0 .. 12)]) :

> # Apply Gram-Schmidt process
  orthoFuncs := GramSchmidt(funcList) :



>
> # Print orthogonalized functions
  psi := map(z→ collect(z, sin(y)), Vector(orthoFuncs)) :
> psi[1 ..9]
```

$$\left[\left[\frac{\sin(y)}{y\sqrt{\pi}}\right], \left[\frac{\sqrt{3}\,\sin(y)}{y^2\sqrt{\pi}} - \frac{\cos(y)\sqrt{3}}{y\sqrt{\pi}}\right], \left[\frac{\sqrt{5}\,(y^2-3)\sin(y)}{y^3\sqrt{\pi}} + \frac{3\sqrt{5}\cos(y)}{y^2\sqrt{\pi}}\right], \left[\frac{(-6y^2+15)\sqrt{7}\,\sin(y)}{\sqrt{\pi}\,y^4} + \frac{(y^3-15y)\cos(y)\sqrt{7}}{\sqrt{\pi}\,y^4}\right], \left[\frac{3\,(y^4-45y^2+105)\sin(y)}{\sqrt{\pi}\,y^5} + \frac{3\,(10y^3-105y)\cos(y)}{\sqrt{\pi}\,y^5}\right], \left[\frac{\sqrt{11}\,(-15y^4+420y^2-945)\sin(y)}{\sqrt{\pi}\,y^6} + \frac{\sqrt{11}\,(y^5-105y^3+945y)\cos(y)}{\sqrt{\pi}\,y^6}\right], \left[\frac{\sqrt{13}\,(y^6-210y^4+4725y^2-10395)\sin(y)}{\sqrt{\pi}\,y^7} + \frac{21\sqrt{13}\cos(y)\,(y^4-60y^2+495)}{\sqrt{\pi}\,y^6}\right], \left[\frac{\sqrt{15}\,(-28y^6+3150y^4-62370y^2+135135)\sin(y)}{\sqrt{\pi}\,y^8} + \frac{\sqrt{15}\,(y^7-378y^5+17325y^3-135135y)\cos(y)}{\sqrt{\pi}\,y^8}\right], \left[\frac{\sqrt{17}\,(y^8-630y^6+51975y^4-945945y^2+2027025)\sin(y)}{\sqrt{\pi}\,y^9} + \frac{36\sqrt{17}\cos(y)\left(y^6-\frac{385}{2}y^4+\frac{15015}{2}y^2-\frac{225225}{4}\right)}{\sqrt{\pi}\,y^8}\right]\right] \tag{4}$$

```
>
> plots[display](seq(plot((-1)^n · psi[2 n]), y = 0.5 ..40, title = n, view = [0.5 ..40, -.5 ..0.5],
    gridlines, axes = boxed), n = 1 ..6))
```



> $plots[display](seq(plot(((-1)^{n+1} \cdot \mathrm{psi}[2\,n-1]), y = 0.5\,..40, title = n, view = [0.5\,..40, -.5\,..0.5], gridlines, axes = boxed), n = 1\,..6))$



>

>

> $hmmsin := map\left(z \rightarrow \left(\frac{op(1, z) \cdot \mathrm{sqrt}(\mathrm{Pi})}{\sin(y)}\right), \mathrm{psi}\right) : coolsin := Vector\left(\left[seq\left(simplify\left(\frac{hmmsin[i] \cdot y^{i}}{\mathrm{sqrt}(2 \cdot i - 1)}\right), i = 1 .. 13\right)\right]\right)$

$coolsin := \left[\left[\sqrt{\pi}\, y\right],\right.$ **(5)**

$$[1],$$
$$[y^2 - 3],$$
$$[-6\,y^2 + 15],$$
$$[y^4 - 45\,y^2 + 105],$$
$$[-15\,y^4 + 420\,y^2 - 945],$$
$$[y^6 - 210\,y^4 + 4725\,y^2 - 10395],$$
$$[-28\,y^6 + 3150\,y^4 - 62370\,y^2 + 135135],$$
$$[y^8 - 630\,y^6 + 51975\,y^4 - 945945\,y^2 + 2027025],$$
$$[-45\,y^8 + 13860\,y^6 - 945945\,y^4 + 16216200\,y^2 - 34459425],$$
$$[y^{10} - 1485\,y^8 + 315315\,y^6 - 18918900\,y^4 + 310134825\,y^2 - 654729075],$$
$$[-66\,y^{10} + 45045\,y^8 - 7567560\,y^6 + 413513100\,y^4 - 6547290750\,y^2 + 13749310575],$$
$$[y^{12} - 3003\,y^{10} + 1351350\,y^8 - 192972780\,y^6 + 9820936125\,y^4 - 151242416325\,y^2 + 316234143225]]$$

> $hmmcos := map\left(z \rightarrow \left(\frac{op(2, z)\cdot \mathrm{sqrt}(\mathrm{Pi})}{\cos(y)}\right), \mathrm{psi}\right) : coolcos :=$
$Vector\left(\left[seq\left(simplify\left(\frac{hmmcos[i]\cdot y^{i-1}}{\mathrm{sqrt}(2\cdot i - 1)}\right), i = 1..13\right)\right]\right)$

$coolcos :=$ **(6)**

$$\left[\begin{array}{c}
\frac{\sqrt{\pi}\,\sec(y)}{y} \\
-1 \\
3 \\
y^2-15 \\
10\,y^2-105 \\
y^4-105\,y^2+945 \\
21\,y^4-1260\,y^2+10395 \\
y^6-378\,y^4+17325\,y^2-135135 \\
36\,y^6-6930\,y^4+270270\,y^2-2027025 \\
y^8-990\,y^6+135135\,y^4-4729725\,y^2+34459425 \\
55\,y^8-25740\,y^6+2837835\,y^4-91891800\,y^2+654729075 \\
y^{10}-2145\,y^8+675675\,y^6-64324260\,y^4+1964187225\,y^2-13749310575 \\
78\,y^{10}-75075\,y^8+18378360\,y^6-1571349780\,y^4+45831035250\,y^2-316234143225
\end{array}\right]$$

> $cosMatrix := convert(convert(map(z \rightarrow ListTools{:}\text{-}Reverse(convert(MTM{:}\text{-}coeffs(z, y), list)), coolcos[3..12]), list), Matrix)$

$$cosMatrix := \left[\begin{array}{cccccc}
3 & 0 & 0 & 0 & 0 & 0 \\
-15 & 1 & 0 & 0 & 0 & 0 \\
-105 & 10 & 0 & 0 & 0 & 0 \\
945 & -105 & 1 & 0 & 0 & 0 \\
10395 & -1260 & 21 & 0 & 0 & 0 \\
-135135 & 17325 & -378 & 1 & 0 & 0 \\
-2027025 & 270270 & -6930 & 36 & 0 & 0 \\
34459425 & -4729725 & 135135 & -990 & 1 & 0 \\
654729075 & -91891800 & 2837835 & -25740 & 55 & 0 \\
-13749310575 & 1964187225 & -64324260 & 675675 & -2145 & 1
\end{array}\right] \quad \textbf{(7)}$$

> $sinMatrix := convert(convert(map(z \rightarrow ListTools{:}\text{-}Reverse(convert(MTM{:}\text{-}coeffs(z, y), list)), coolsin[3..12]), list), Matrix)$

(8)

$$sinMatrix := \begin{bmatrix} -3 & 1 & 0 & 0 & 0 & 0 \\ 15 & -6 & 0 & 0 & 0 & 0 \\ 105 & -45 & 1 & 0 & 0 & 0 \\ -945 & 420 & -15 & 0 & 0 & 0 \\ -10395 & 4725 & -210 & 1 & 0 & 0 \\ 135135 & -62370 & 3150 & -28 & 0 & 0 \\ 2027025 & -945945 & 51975 & -630 & 1 & 0 \\ -34459425 & 16216200 & -945945 & 13860 & -45 & 0 \\ -654729075 & 310134825 & -18918900 & 315315 & -1485 & 1 \\ 13749310575 & -6547290750 & 413513100 & -7567560 & 45045 & -66 \end{bmatrix} \quad (8)$$

> $B := (n, k) \rightarrow \text{pochhammer}(n+1, k) * \text{binomial}(n, k) / 2\text{^}k$

$$B := (n, k) \mapsto \frac{\text{pochhammer}(n+1, k) \cdot \binom{n}{k}}{2^k} \quad (9)$$

> $C := unapply(simplify(\ (-1)^{n+1} \cdot (-1)\text{^}\text{binomial}(k, 2) \cdot B(k+1, k-(2n-1))), n, k)$ assuming $n :: nonnegint, k :: nonnegint$

$$C := (n, k) \mapsto -(-1)^{n+\binom{k}{2}} \cdot 2^{-1-k+2\cdot n} \cdot \binom{k+1}{k-2\cdot n+1} \cdot \text{pochhammer}(k+2, k-2\cdot n+1) \quad (10)$$

>

>

> $convert([seq(flist(C(n, k), n = 0..k), k = 1..10)], Matrix);$
first col is constants, 2nd col is coeffs of x, 3rd is of x^2 ...

$$\begin{bmatrix} -3 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 15 & -6 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 105 & -45 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -945 & 420 & -15 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -10395 & 4725 & -210 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 135135 & -62370 & 3150 & -28 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 2027025 & -945945 & 51975 & -630 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ -34459425 & 16216200 & -945945 & 13860 & -45 & 0 & 0 & 0 & 0 & 0 & 0 \\ -654729075 & 310134825 & -18918900 & 315315 & -1485 & 1 & 0 & 0 & 0 & 0 & 0 \\ 13749310575 & -6547290750 & 413513100 & -7567560 & 45045 & -66 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad (11)$$

> $sinMatrix$

$$\begin{bmatrix} -3 & 1 & 0 & 0 & 0 & 0 \\ 15 & -6 & 0 & 0 & 0 & 0 \\ 105 & -45 & 1 & 0 & 0 & 0 \\ -945 & 420 & -15 & 0 & 0 & 0 \\ -10395 & 4725 & -210 & 1 & 0 & 0 \\ 135135 & -62370 & 3150 & -28 & 0 & 0 \\ 2027025 & -945945 & 51975 & -630 & 1 & 0 \\ -34459425 & 16216200 & -945945 & 13860 & -45 & 0 \\ -654729075 & 310134825 & -18918900 & 315315 & -1485 & 1 \\ 13749310575 & -6547290750 & 413513100 & -7567560 & 45045 & -66 \end{bmatrix} \tag{12}$$

```
> # Now to find the function for the cosMatrix
> cosMatrix
```

$$\begin{bmatrix} 3 & 0 & 0 & 0 & 0 & 0 \\ -15 & 1 & 0 & 0 & 0 & 0 \\ -105 & 10 & 0 & 0 & 0 & 0 \\ 945 & -105 & 1 & 0 & 0 & 0 \\ 10395 & -1260 & 21 & 0 & 0 & 0 \\ -135135 & 17325 & -378 & 1 & 0 & 0 \\ -2027025 & 270270 & -6930 & 36 & 0 & 0 \\ 34459425 & -4729725 & 135135 & -990 & 1 & 0 \\ 654729075 & -91891800 & 2837835 & -25740 & 55 & 0 \\ -13749310575 & 1964187225 & -64324260 & 675675 & -2145 & 1 \end{bmatrix} \tag{13}$$

```
>
> #  See Juggling Zeros in the Matrix (Example II)
  # Tom Copeland, Cheviot Hills, Los Angeles, CA, July 15-17, 20270
>
```